Cancer among Hispanic Women in South Florida: An 18-Year Assessment

A Report from the Florida Cancer Data System

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BACKGROUND. The Hispanic population now represents the majority of residents in Miami-Dade County, Florida. The authors present cancer incidence and mortality data for South Florida's Hispanic women for the period 1990–1998 and compare these data to previously reported data from 1981–1989. Cancer incidence, risk, and mortality data should reflect current population distribution, lifestyle, and environmental risk factors so that cancer prevention and control activities are informed optimally.

METHODS. The study population consisted of all women with malignant disease during 1981–1998 from Miami-Dade County found in the Florida Cancer Data System data base; patients were divided into 2 9-year periods for analysis. Age-standardized incidence and mortality rates were computed for common disease sites; rates for Hispanic women were compared with the rates for non-Hispanic white (NHW) women as standardized rate ratios (SRR) with 95% confidence intervals (95%CIs). Incidence and mortality trends were analyzed using linear regression.

RESULTS. Over 70,000 cancer incidents were analyzed. The overall decreased cancer risk for Hispanic women (SRR, 0.65; 95%CI, 0.64–0.67), compared with NHW women, remained essentially constant over the two study periods. Cancer incidence increased similarly for the two racial-ethnic groups. The incidence of lung carcinoma increased in both groups, becoming the second most common disease site for NHW women and the third most common disease site for Hispanic women.

CONCLUSIONS. The decreased relative cancer risk for Hispanic women in South Florida has remained stable over the past 18 years. Lung carcinoma is increasing among women in both racial-ethnic groups. ***Cancer*** **2002;95:1752–8.**

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Many of the malignant diseases of greatest significance to women (such as carcinoma of the breast, colorectal, and uterine cervix) are amenable to effective cancer prevention and control strategies. These strategies include lifestyle changes, cancer screening participation, and health risk assessment. Current and accurate information regarding cancer incidence and risk are essential to inform best practices with regard to cancer prevention and control.

The Hispanic population is the fastest growing minority population in the United States and represents the majority of residents in Miami-Dade County, Florida. The Hispanic population in South Flor-

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ida is primarily of Cuban national origin. In general, the tumor burden for Hispanic patients in the United States has been less compared with the tumor burden for non-Hispanic white (NHW) patients in the United States. However, it may be expected that cancer incidence rates among the Hispanic population will increase due to changing exposures associated with increased acculturation. It has been demonstrated that incidence rates for some common malignancies (lung, breast, and prostate carcinoma) may vary by as much as 40–50% among Hispanic populations from different regions of the United States.[1] To date, most reports of cancer in Hispanic populations have concerned persons of Mexican or Puerto Rican ancestry. It is important to note that published data regarding cancer in Hispanics reflect the diversity of this population.

We previously reported cancer incidence data for Hispanic women in South Florida during the period 1981–1989.[2,3] Now, we present new cancer incidence data for South Florida Hispanic women for the period 1990–1998, including temporal trends for some of the more common malignancies for the period 1981–1998. Current and accurate cancer surveillance data are vital to the development of effective and targeted cancer prevention and control strategies. The examination of ethnic differences in cancer incidence and trends is consistent with the Institute of Medicine's recent recommendation for an increased emphasis on ethnic groups (cultural, behavioral, and lifestyle differences) rather than on race (biologic differences) in cancer surveillance.[4]

MATERIALS AND METHODS

This study was performed in collaboration with the Florida Cancer Data System (FCDS), the State of Florida population-based cancer registry that was established in 1981, and was supported by the Florida Department of Health and the Centers for Disease Control and Prevention (CDC). The methods described below have been reported in greater detail previously.[5]

The study population consisted of all incidence of cancer occurring in women from Miami-Dade County, Florida, whose race or ethnicity was classified as either *NHW* (the reference group) or *Hispanic*, regardless of race. Less than 10% of the FCDS population is designated both as a race other than white and Hispanic; these *NHW* women were included in the racial-ethnic group of *Hispanic* women for study purposes. Patients for whom Hispanic ethnicity was unknown ($< 3\%$) were excluded from further analysis. Patients were then divided into two 9-year periods for analysis: 1981–1989 (early period) and 1990–1998 (late period).

Reported disease sites were selected to represent the most common and important malignancies in women and to be consistent with reports from other investigators. Specific disease sites according to International Classification of Diseases for Oncology codes were grouped according to North American Association of Central Cancer Registries (NAACCR) standards. Breast carcinoma was analyzed as a disease site, with two subgroups: patients with in situ breast carcinoma and patients with invasive breast carcinoma. In this report, references to cervical carcinoma apply only to patients with invasive disease, not in situ disease, consistent with CDC guidelines that only invasive cervical carcinoma is tracked. Although melanoma is an important malignancy in Florida, it was not reported in this study as a result of concerns by FCDS personnel related to known inconsistent and incomplete reporting on patients with melanoma.

Results were calculated and are expressed as age-standardized incidence rates using the 1970 United States standard million population and Florida intercensus population projections for the racial-ethnic groups defined above. The probability of acquiring cancer for a racial-ethnic group was expressed as the standardized rate ratio (SRR) with the two-sided 95% confidence interval (95%CI) using NHW women as the reference group. Incidence and mortality trends were analyzed using linear regression based on 3-year moving averages for the years for which complete racial-ethnic data were available. Statistical significance for the regressions was set at $\alpha = 0.05$. The data were analyzed using SAS software (version 8.0; SAS Institute, Cary, NC) and were exported to Microsoft Excel 2000 software (Microsoft Corp., Redmond, WA) for presentation purposes.

RESULTS

From 1981 to 1998, 70,017 incidents of cancer were identified among NHW women and Hispanic women in Miami-Dade County, Florida. Consistent with demographic changes in the county's population, Hispanic women accounted for 36% of the incidents of cancer during the earlier study period and accounted for 52% of the incidents of cancer during the later period.

Cancer Incidence: Age-Adjusted Rates and SRRs

The cancer incidence (age adjusted rates per 10^5 population) for each disease site by race-ethnicity for the two study periods is shown in Table 1. The overall incidence for NHW women increased 13%, from 383 to 432 per 100,000 person-years, across the 2 study periods. The rate for Hispanic women increased 16%,

TABLE 1
Cancer Incidence for Non-Hispanic White Women and Hispanic Women in Miami-Dade County, Florida: 1981–1998

	1981–1989				1990–1998			
Site	Hispanic patients	Hispanic rates[a]	NHW rates[a]	NHW patients	Hispanic patients	Hispanic rates[a]	NHW rates[a]	NHW patients
Head and neck	182	3.7 (0.3)	8.8 (0.5)	424	283	3.8 (0.2)	8.4 (0.5)	332
Esophagus	36	0.7 (0.1)	1.8 (0.2)	118	69	0.8 (0.1)	2.1 (0.2)	99
Stomach	212	4.2 (0.3)	5.4 (0.3)	394	372	4.6 (0.3)	4.3 (0.3)	238
Colorectal	1962	39.2 (0.9)	50.1 (1.0)	3413	3093	40.3 (0.8)	47.6 (1.1)	2550
Liver	91	1.9 (0.2)	1.2 (0.2)	72	156	2.0 (0.2)	1.5 (0.2)	73
Gall bladder	110	2.1 (0.2)	1.4 (0.1)	85	182	2.4 (0.2)	0.8 (0.1)	43
Pancreas	242	4.7 (0.3)	7.5 (0.4)	532	487	6.0 (0.3)	8.4 (0.5)	469
Lung and bronchus	599	12.0 (0.5)	47.2 (1.0)	2623	1245	17.3 (0.5)	52.0 (1.2)	2327
Breast[b]	3555	75.7 (1.3)	120.7 (1.8)	5946	6077	92.2 (1.2)	144.9 (2.1)	5458
Breast, in situ	134	2.9 (0.3)	7.5 (0.5)	316	641	10.3 (0.4)	19.1 (0.8)	651
Breast, invasive	3421	72.8 (1.3)	113.2 (1.7)	5630	5436	81.9 (1.2)	125.8 (2.0)	4807
Cervix	473	10.3 (0.5)	10.3 (0.6)	411	775	12.2 (0.4)	11.5 (0.7)	358
Corpus uteri	771	16.1 (0.6)	20.2 (0.7)	1082	1134	16.8 (0.5)	23.1 (0.8)	925
Ovary	556	12.1 (0.5)	17.7 (0.7)	874	781	12.0 (0.4)	19.4 (0.8)	710
Vagina	32	0.7 (0.1)	1.1 (0.2)	57	51	0.7 (0.1)	1.1 (0.2)	48
Vulva	73	1.5 (0.2)	4.0 (0.3)	189	127	1.7 (0.2)	4.7 (0.4)	178
Urinary bladder	243	4.7 (0.3)	8.7 (0.4)	596	334	3.9 (0.2)	8.7 (0.5)	472
Kidney	186	4.0 (0.3)	5.9 (0.4)	339	354	5.2 (0.3)	7.1 (0.5)	308
Brain	167	4.1 (0.3)	4.9 (0.4)	203	293	5.2 (0.3)	5.7 (0.5)	191
Thyroid	246	5.5 (0.4)	5.1 (0.4)	197	378	6.3 (0.3)	9.1 (0.6)	247
Hodgkin lymphoma	74	1.9 (0.2)	2.3 (0.3)	82	114	2.2 (0.2)	3.0 (0.4)	64
Non-Hodgkin lymphoma	363	7.5 (0.4)	10.3 (0.5)	619	714	10.5 (0.4)	13.3 (0.6)	602
All leukemias	239	5.7 (0.4)	8.0 (0.5)	426	448	7.1 (0.4)	8.3 (0.6)	361
All sites	11,569	242.6 (2.3)	382.3 (3.0)	20,885	19,523	282.8 (2.1)	431.8 (3.7)	18,040

NHW: non-Hispanic white.
[a] Per 100,000 person-years. Values in parentheses are the standard error.
[b] Includes in situ and invasive carcinoma.

from 243 to 283 per 100,000 person-years over the same periods.

With regard to risk, the SRRs for all disease sites for Hispanic women, compared with NHW women, were 0.63 and 0.65, respectively. Thus, there was no appreciable change in the overall decreased cancer risk for Hispanic women over the two study periods.

During the earlier period, 1981–1989, breast and colorectal carcinoma were the most common malignancies in women from both racial-ethnic groups, with carcinoma of the lung, uterus, and ovary among the top five most common, although in different order for the two racial-ethnic groups (Table 2). During the more recent period, lung carcinoma moved to second place for NHW women and moved to third place for Hispanic women. For NHW women, colorectal carcinoma, carcinoma of the corpus uteri, and ovarian carcinoma completed the top five incidence list. For Hispanic women, colorectal carcinoma remained in second place, carcinoma of the corpus uteri moved to fourth place, and cervical carcinoma moved onto the top five list, replacing ovarian carcinoma.

TABLE 2
Five Most Common Cancer Sites for Non-Hispanic White Women and Hispanic Women in Miami-Dade County, Florida: 1981–1989 and 1990–1998

	1981–1989		1990–1998	
Rank	NHW	Hispanic	NHW	Hispanic
1	Breast	Breast	Breast	Breast
2	Colorectal	Colorectal	Lung	Colorectal
3	Lung	Corpus uteri	Colorectal	Lung
4	Corpus uteri	Ovary	Corpus uteri	Corpus uteri
5	Ovary	Lung	Ovary	Ovary

NHW: non-Hispanic white.

Female Reproductive Malignancies

The incidence of total breast carcinomas increased about 20% for both NHW women and Hispanic women over the study periods. Approximately equal increases in the incidence of both in situ disease and invasive disease accounted for the overall increased incidence for both groups (Table 1). The decreased

TABLE 3
Standardized Rate Ratios with 95% Confidence Intervals for Common Cancer Sites among Women in Miami-Dade County, Florida: 1981–1989 and 1990–1998

	1981–1989		1990–1998	
Site	SRR (Hispanic: NHW)	95%CI	SRR (Hispanic: NHW)	95%CI
Head and neck	0.42	0.35–0.51	0.45	0.37–0.54
Esophagus	0.40	0.28–0.57	0.38	0.26–0.56
Stomach	0.78	0.65–0.92	1.07	0.89–1.29
Colorectal	0.78	0.74–0.83	0.85	0.80–0.90
Liver	1.60	1.15–2.22	1.31	0.97–1.78
Gall bladder	1.87	1.38–2.55	3.16	2.30–4.35
Pancreas	0.63	0.54–0.73	0.71	0.61–0.83
Lung and bronchus	0.26	0.23–0.28	0.33	0.31–0.36
Breast[a]	0.63	0.60–0.66	0.64	0.61–0.66
Breast, in situ	0.39	0.31–0.48	0.54	0.48–0.61
Breast, invasive	0.64	0.61–0.67	0.65	0.62–0.68
Cervix	1.00	0.87–1.15	1.05	0.93–1.20
Corpus uteri	0.80	0.72–0.88	0.73	0.66–0.80
Ovary	0.68	0.61–0.76	0.62	0.55–0.69
Vagina	0.60	0.38–0.96	0.62	0.39–0.99
Vulva	0.37	0.28–0.50	0.36	0.27–0.48
Urinary bladder	0.54	0.47–0.63	0.44	0.37–0.53
Kidney	0.68	0.56–0.83	0.73	0.61–0.87
Brain	0.83	0.66–1.05	0.92	0.75–1.14
Thyroid	1.07	0.88–1.30	0.69	0.57–0.83
Hodgkin lymphoma	0.81	0.58–1.15	0.73	0.50–1.04
Non-Hodgkin lymphoma	0.73	0.63–0.83	0.79	0.70–0.90
All leukemias	0.72	0.60–0.86	0.85	0.72–1.01
All sites	0.63	0.62–0.65	0.65	0.64–0.67

SRR: standardized rate ratio; NHW: non-Hispanic white; 95%CI: 95% confidence interval.
[a] Includes in situ and invasive carcinoma.

relative risk for Hispanic women for both total breast carcinomas and invasive disease was similar over the two study periods, with an SRR of 0.63 (95%CI, 0.60–0.66). The relative risk for in situ breast carcinoma among Hispanic women did increase over the study periods from 0.4 (95%CI, 0.31–0.48) to 0.54 (95%CI, 0.48–0.61) (Table 3). With regard to trends, the incidence of total breast carcinomas and both site subgroups increased significantly over the study periods for both NHW women and Hispanic women, with a rate of increase the same or slightly higher for NHW women (Table 4).

For carcinoma of the uterus (corpus uteri), there were increases in incidence over the two study periods for both racial-ethnic groups: from 20 to 23 per 100,000 person-years for NHW women and from 16 to nearly 17 per 100,000 person-years for Hispanic women. (Table 1) The SRR for corpus uteri carcinoma in Hispanic women during the earlier study period was 0.8 and decreased slightly to 0.73 during the later study period. The incidence increase was significant for NHW women but not for Hispanic women (Table 4).

There was little change in either incidence or SRR for ovarian carcinoma over the two study periods. The SRRs for Hispanic women for ovarian carcinoma were 0.68 and 0.62 for the early and late study periods, respectively (Table 3). There was no significant incidence trend for ovarian carcinoma for either racial-ethnic group (Table 4).

For invasive cervical carcinoma, the incidence rates in the early study period were identical (10.3 per 100,000 person-years) for the two racial-ethnic groups, with small increases resulting in a slightly higher incidence for Hispanic women compared with NHW women, in the later study period (Table 1). The increasing incidence trend was only significant for Hispanic women (Table 4). For both the early and late study periods, the SRR for invasive cervical carcinoma was the same for both racial-ethnic groups (Table 3).

Nonreproductive Disease Sites

Lung carcinoma continued to be more than three times as common among NHW women over both study periods. The most recent rate among NHW women was 52 per 100,000 person-years, with a rate among Hispanic women of 17 per 100,000 person-years. The associated SRR increased from 0.26 (95%CI, 0.23–0.28) to 0.33 (95%CI, 0.31–0.36) over the study periods. With regard to trends, the increase in the incidence of lung carcinoma among NHW women was barely significant ($P = 0.04$); however, there was a strongly significant increase among Hispanic women ($P < 0.0001$) over the study periods. It is noteworthy that lung carcinoma has displaced colorectal carcinoma as the second most common malignancy among NHW women and moved from fifth place to third place among Hispanic women over the study periods.

Colorectal carcinoma was the most important gastrointestinal malignancy for both racial-ethnic groups, with incidence rates for both groups of approximately 40–50 per 100,000 person-years. The SRR for Hispanic women was stable over the study periods; 0.78 (95%CI, 0.74–0.83) and 0.85 (95%CI, 0.80–0.90) for the early and late study periods, respectively. There was no demonstrable incidence trend for NHW women over the two study periods; however, for Hispanic women, there was a significant increase ($P = 0.03$).

In the earlier study, hepatocellularcarcinoma was one of the only malignancies that showed a significantly increased risk for Hispanic women compared with NHW women, with an SRR of 1.6 (95%CI, 1.15–

TABLE 4
Cancer Site Incidence Trends for Women in Miami-Dade County, Florida: Linear Regression Using 3-Year Moving Averages, 1981–1998

	Hispanic			NHW		
Site	β_1	*P* value	95%CI	β_1	*P* value	95%CI
Head and neck	− 0.03	0.22	− 0.08–0.02	0.00	0.91	− 0.07–0.06
Esophagus	0.02	0.04	0.00–0.04	0.01	0.52	− 0.02–0.03
Stomach	0.03	0.21	− 0.02–0.08	− 0.05	0.16	− 0.11–0.02
Colorectal	0.31	0.03	0.04–0.59	− 0.20	0.22	− 0.54–0.14
Liver	0.01	0.68	− 0.04–0.06	0.07	0.00	0.04–0.10
Gall bladder	0.02	0.25	− 0.02–0.06	− 0.04	0.01	− 0.08 to − 0.01
Pancreas	0.15	0.00	0.08–0.22	0.12	0.01	0.04–0.20
Lung and bronchus	0.68	0.00	0.57–0.79	0.48	0.04	0.02–0.95
Breast[a]	2.13	0.00	1.82–2.43	2.94	0.00	2.09–3.80
Breast, in situ	0.80	0.00	0.70–0.90	1.41	0.00	1.09–1.73
Breast, invasive	1.33	0.00	1.01–1.65	1.54	0.00	0.85–2.22
Cervix	0.18	0.00	0.13–0.24	0.08	0.21	− 0.05–0.21
Corpus uteri	0.10	0.06	− 0.01–0.21	0.33	0.00	0.17–0.49
Ovary	0.04	0.47	− 0.07–0.15	0.23	0.05	0.00–0.46
Vagina	0.00	0.73	−0.02–0.02	0.02	0.07	0.00–0.04
Vulva	0.05	0.01	0.01–0.08	0.11	0.00	0.04–0.18
Urinary bladder	− 0.06	0.06	− 0.12–0.00	0.07	0.10	− 0.01–0.16
Kidney	0.13	0.01	0.04–0.22	0.17	0.00	0.09–0.24
Brain	0.11	0.00	0.07–0.14	0.07	0.15	− 0.03–0.16
Thyroid	0.13	0.00	0.07–0.20	0.49	0.00	0.25–0.72
Hodgkin lymphoma	0.03	0.05	0.00–0.06	0.07	0.01	0.02–0.13
Non-Hodgkin lymphoma	0.33	0.00	0.25–0.42	0.37	0.00	0.29–0.45
All leukemias	0.14	0.00	0.07–0.21	0.05	0.22	− 0.03–0.12
All sites	5.16	0.00	4.36–5.97	6.23	0.00	4.25–8.22

NHW: non-Hispanic whites; 95%CI: 95% confidence interval.
[a] Includes in situ and invasive carcinoma.

2.22). Although the SRR remained elevated at 1.3 in the later study period, it was no longer statistically significant (95%CI, 0.97–1.78). Carcinoma of the gall bladder had an increased risk for Hispanic women that persisted over both study periods, with SRRs of 1.9 (95%CI, 1.38–2.55) and 3.2 (95%CI, 2.30–4.35), respectively. Nevertheless, this malignancy was relatively rare, with 100–200 diagnoses among Hispanic women for each 9-year period.

Cancer Mortality Trends

Overall cancer mortality decreased significantly over both study periods for both NHW and Hispanic women, with mortality decreasing more sharply among the former (Table 5). In 1998, the mortality rate for NHW women was 124 per 100,000 person-years, and the mortality rate for Hispanic women was 90 per 100,000 person-years.

Female Reproductive Malignancies

For breast carcinoma, mortality decreased for both racial-ethnic groups, with a sharper decline seen among NHW women (Table 5). Mortality from carcinoma of the corpus uteri decreased significantly among NHW women but showed a small but significant increase among Hispanic women. Mortality from ovarian carcinoma decreased significantly for NHW women over both study periods, but there was no significant mortality trend identified among Hispanic women. The mortality trend for cervical carcinoma was flat among NHW women but showed a small but significant, increase among Hispanic women over both study periods.

Nonreproductive Disease Sites

Lung carcinoma mortality increased among Hispanic women but decreased among NHW women. Among digestive malignancies, mortality decreased for colorectal, liver, and gall bladder carcinoma among Hispanic women only; gastric and pancreatic carcinoma decreased only among NHW women. Regarding lymphomas, mortality for Hodgkin lymphoma decreased for Hispanic women and mortality for non-Hodgkin lymphoma decreased for NHW women.

TABLE 5
Cancer Site Mortality Trends for Women in Miami-Dade County, Florida: Linear Regression Using 3-Year Moving Averages, 1989–1998[a]

	Hispanic			NHW		
Site	β_1	*P* value	95%CI	β_1	*P* value	95%CI
Head and neck	− 0.04	0.04	− 0.07–0.00	− 0.06	0.18	− 0.16–0.04
Esophagus	0.03	0.18	− 0.02–0.08	− 0.02	0.30	− 0.06–0.02
Stomach	− 0.01	0.72	− 0.06–0.04	− 0.29	0.00	− 0.38 to − 0.20
Colorectal	− 0.30	0.01	− 0.48 to − 0.12	− 0.55	0.06	− 1.13–0.03
Liver	− 0.06	0.04	− 0.12–0.00	− 0.07	0.06	− 0.14–0.00
Gall bladder	− 0.06	0.00	− 0.09 to − 0.03	− 0.02	0.14	− 0.04–0.01
Pancreas	0.06	0.40	− 0.09–0.21	− 0.29	0.02	− 0.51 to − 0.06
Lung and bronchus	0.12	0.05	0.00–0.23	− 1.07	0.01	− 1.72 to − 0.41
Breast	− 0.30	0.02	− 0.53 to − 0.06	− 1.39	0.00	− 1.68 to − 1.10
Cervix	0.10	0.00	0.05–0.15	− 0.01	0.82	− 0.06–0.05
Corpus uteri	0.04	0.01	0.01–0.07	− 0.09	0.03	− 0.16 to − 0.01
Ovary	− 0.09	0.05	− 0.19–0.00	− 0.31	0.00	− 0.46 to − 0.17
Vagina	− 0.02	0.01	− 0.03 to − 0.01	0.03	0.00	0.02–0.04
Vulva	− 0.01	0.03	− 0.02–0.00	− 0.02	0.05	0.05–0.00
Urinary bladder	− 0.05	0.17	− 0.12–0.03	− 0.04	0.22	− 0.12–0.03
Kidney	− 0.08	0.02	− 0.14 to − 0.02	− 0.05	0.09	− 0.10–0.01
Brain	− 0.01	0.78	− 0.13–0.11	− 0.16	0.04	− 0.32 to − 0.01
Thyroid	0.00	0.59	− 0.01–0.02	− 0.04	0.01	− 0.07 to − 0.01
Hodgkin lymphoma	− 0.04	0.02	− 0.06 to − 0.01	0.00	0.98	− 0.02–0.02
Non-Hodgkin lymphoma	0.10	0.09	− 0.02–0.22	− 0.33	0.04	− 0.64 to − 0.01
Leukemias	0.04	0.31	− 0.05–0.13	− 0.09	0.11	− 0.21–0.03
All sites	− 0.66	0.02	− 1.16 to − 0.16	− 5.67	0.00	− 8.00 to − 3.33

NHW: non-Hispanic whites; 95%CI: 95% confidence interval.
[a] Mortality rates for Hispanic ethnicity were not available before 1989.

DISCUSSION

The incidence of cancer among women in Miami-Dade County, Florida, increased over the two study periods in a similar manner for both racial-ethnic groups: about 13% for NHW women and 16% for Hispanic women. With regard to breast carcinoma, the proportion of patients with in situ disease more than doubled over both study periods for both groups, whereas mortality has been decreasing. This may represent improvements in awareness and use of screening procedures, such as mammography. During the late 1980s, the University of Miami started a community-based breast cancer early detection program, with special emphasis placed on Hispanic women, although it is not expected that this alone accounts for the difference.[6–8]

Regarding SRRs for reproductive malignancies, the risk for Hispanic women of breast (invasive), cervical, corpus uteri, and ovarian carcinomas have remained fairly stable over the study periods, with a risk for all but cervical carcinoma that is 20–40% less compared with the risk for NHW women. The SRR for cervical carcinoma is approximately 1.0 and is due to an increased rate for NHW women (compared with the reported national SEER rates), compared with a decreased risk for Hispanic women.[9]

In terms of both incidence and trends, the reproductive malignances in South Florida women are fairly comparable to the national experience from the SEER registry with two exceptions:[9] The incidence of breast carcinoma in Miami-Dade Hispanic women is increasing, whereas the national trend is nearly flat. The rate for cervical carcinoma among non-Hispanic women in Miami-Dade County, as noted above, is nearly 50% higher compared with the national SEER rate. This may reflect a higher rate of sexually transmitted diseases in a mostly urban, relatively poor county, as evidenced by the highest rates of human immunodeficiency virus and acquired immunodeficiency syndrome in the nation.[10]

Lung carcinoma rates in Miami-Dade County are comparable to the national SEER rates, but the trends are not. National trends for both non-Hispanic women and Hispanic women are decreasing slightly, whereas they continue to increase between both of these groups, particularly for Hispanic women, in Miami-Dade County. Colon carcinoma rates are nearly twice as high among Hispanic women in Miami-Dade

County compared with the national SEER rates for Hispanic women. These trends may reflect differences between the primarily Cuban and Central American national origin of the Hispanic population in South Florida compared with primarily Mexican and Puerto Rican populations elsewhere in the United States as well as differences in assimilation regarding the acceptance of cigarette use by women, dietary differences, and/or other yet unidentified factors.

With regard to study limitations, as discussed in our previous article on cancer incidence in Hispanic men, misclassification of ethnicity is always a potential methodological problem.[5,11,12] However, ongoing quality assessments indicate that FCDS ethnicity data, particularly for Miami-Dade County, are quite accurate. Other potential limitations also should be considered. Regarding comparisons of incidence rates, the only variables controlled for were age, gender, and major racial-ethnic group. Other pertinent variables, particularly the amount of acculturation and genetic differences, are not addressed in the FCDS data base and, thus, were not addressed in this study. Finally, statistically significant differences should be interpreted cautiously, because they may reflect both true differences as well as the large size of the populations being compared.

Differences in cancer risk among Hispanic women may represent differences in both genetic and environmental exposures, although migrant studies have demonstrated that environmental factors dominate in the epidemiology of many malignancies, including breast and colon carcinoma.[13] Studies have shown differences among Hispanics concerning diet, tobacco use, socioeconomic class, attitudes toward serious illness (particularly malignant disease), psychosocial needs, and participation in screening and other health-promoting activities.[14–18]

With the increased acculturation associated with longer residence in the United States, it may be expected that the cancer risk of Hispanic women in South Florida would approximate more closely the risk of native-born non-Hispanic whites women. However, the current findings indicate that important differences in cancer risk persist between non-Hispanic women and Hispanic women in Miami-Dade County. Unfortunately, the FCDS data set does not include length of residence in the United States; thus, we were unable to use that proxy for acculturation in our analysis. Hispanics are the fastest growing ethnic group in the United States, and a better understanding of their cancer burden is important to public health efforts. Because so many important malignancies that affect women are amenable to prevention and control strategies, appreciation of these differences is important for appropriately informed and targeted cancer control and prevention efforts in this and other Hispanic communities in the 21st century.

REFERENCES

1. Trapido EJ, Valdez RB, Obeso JL, Stickman-Stein N, Rotger A, Perez-Stable E. Epidemiology of cancer among Hispanics in the United States. *J Natl Cancer Inst Monogr.* 1995;18:17–28.
2. Trapido EJ, McCoy CB, Stein NS, Engel S, McCoy HV, Olejniczak S. The epidemiology of cancer among Hispanic women: the experience in Florida. *Cancer.* 1990;66:2435–2441.
3. Trapido EJ, Chen F, Davis K, Lewis N, MacKinnon JA, Strait PM. Cancer in South Florida Hispanic women. *Arch Intern Med.* 1994;154:1083–1088.
4. Haynes AM, Smedley BD, editors. The unequal burden of cancer: an assessment of National Institutes of Health research and programs for ethnic minorities and the medically underserved. The Committee on Cancer Research Among Minorities and the Medically Underserved, Institute of Medicine. Washington, DC: National Academy Press, 1999.
5. Wilkinson JD, Wohler-Torres B, Trapido E, et al. Cancer trends among Hispanic Men in South Florida: 1981–1998. *Cancer.* 2002;94:1183–1190.
6. McCoy CB, Nielsen BB, Chitwood DD, Zavertnik JJ, Khoury EL. Increasing the cancer screening of the medically underserved in south Florida. *Cancer.* 1991;67(6 Suppl):1808–1813.
7. Nielsen B, Miaskowski C, McCoy C, Rudisch M. The development and implementation of standards of care in a breast cancer screening program. *Oncol Nurs Forum.* 1991;18:67–72.
8. Zavertnik JJ, McCoy CB, Robinson DS, Love N. Cost-effective management of breast cancer. *Cancer.* 1992;69(7 Suppl): 1979–1984.
9. National Cancer Institute. Canques resources page [Available from URL: http://canques.seer.cancer.gov]. Bethesda: National Cancer Institute.
10. Centers for Disease Control and Prevention. Division of HIV/AIDS Prevention: cases through June 2001 [Available from URL: http://www.cdc.gov/hiv/stats/hasr1301/table4.htm]. Atlanta: Centers for Disease Control and Prevention.
11. Lin SS, Kelsey JL. Use of race and ethnicity in epidemiologic research: concepts, methodological issues, and suggestions for research. *Epidemiol Rev.* 2000;22:187–201.
12. Winkleby MA, Rockhill B. Comparability of self-reported Hispanic ethnicity and Spanish surname coding. *Hispanic J Behav Sci.* 1992;14:487–495.
13. Colditz GA. Cancer culture: epidemics, human behavior, and the dubious search for new risk factors. *Am J Public Health.* 2001;91:357–359.
14. Mitchell JL. Cross-cultural issues in the disclosure of cancer. *Cancer Pract.* 1998;6:153–‘60.
15. Moadel A, Morgan C, Fatone A, et al. Seeking meaning and hope: self-report spiritual and existential needs among an ethnically diverse cancer patient population. *Psychooncology.* 1999;8:378–385.
16. Nichaman MZ, Garcia G. Oesity in Hispanic Americans. *Diabetes Care.* 1991;7:691–694.
17. Perez-Stable EJ, Sabogal F, Otero-Sabogal R, Hiatt RA, McPhee SJ. Misconceptions about cancer among Latinos and Anglos. *JAMA.* 1992;268:3219–3223.
18. Spencer SM, Lehman JM, Wynings C, et al. Concerns about breast cancer and relations to psychosocial well-being in a multiethnic sample of early-stage patients. *Health Psychol.* 1999;18:159–168.